not be periodic and the numerical method then computes a discrete sequence of points which gradually fill in the ellipse (see the right panel of Fig. 3.1).

Obviously this conservation property (the existence of a perturbed energy surface) is a special feature of the method we have considered. If we used Euler's method to solve the harmonic oscillator we would find that energy grows without bound.

Now consider the more general Hamiltonian setting. Let $\mathscr{G}_h$ be a rth order symplectic integrator, $r \geq 1$. Suppose that it is the flow map of a certain Hamiltonian system, with Hamiltonian $\tilde{H}_h$. If the method order is r, we may expect this Hamiltonian to be a $\mathcal{O}(h^r)$ approximation of H, thus we posit an expansion of the form

$$\tilde{H}_h = H + h^r H^{(r)} + h^{r+1} H^{(r+1)} + \dots$$

To determine the terms $H^{(r)}, H^{(r+1)}, \dots$, we write the differential equations on $\tilde{H}_h$:

$$\dot{z} = \boldsymbol{J}\nabla H + h^r \boldsymbol{J}\nabla H^{(r)} + h^{r+1}\boldsymbol{J}\nabla H^{(r+1)} + \dots$$

The solution of this system can be expanded in powers of h and equated term-by-term with the expansion of $\mathscr{G}_h$ in powers of h. In this way, successive terms may be computed. Although mechanical, this procedure is tedious. For splitting methods, a more elegant approach is available using the *Lie derivative*, which we next introduce.

3.2 Lie Derivatives and Poisson Brackets

Let $\phi(z)$ be an arbitrary smooth, scalar-valued function of the phase variables (assumed to lie in $\mathbb{R}^m$). At any point $z = \zeta$ in phase space assume that there is a unique solution $z(t, \zeta)$ such that $\dot{z}(t) = \boldsymbol{f}(z(t)), z(0) = \zeta$ that is globally defined for all t. Now differentiate $\hat{\phi}(t) = \phi(z(t))$ with respect to time, using the chain rule, to see how $\phi(z(t))$ is changing as t is varied:

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\hat{\phi}\right|_{t=0} = \left[\sum_{i=1}^{m} \frac{\partial \phi}{\partial z_i}\dot{z}_i\right]\Bigg|_{z=\zeta} = \nabla\phi(\zeta)\cdot\boldsymbol{f}(\zeta).$$

The right hand side suggests a shorthand notation. If we define the Lie derivative $\mathcal{L}_f$ by

$$\mathcal{L}_f\phi = \boldsymbol{f}\cdot\nabla\phi,$$